

PHASE 1 — FOUNDATION

Week 6 — Control Box II

Finish the wiring. Build the power. Meter-check before the smoke.

Say Watt Robotics · ROV Team | 2026-2027 MATE ROV Season

MEETING 6 · THREE HOURS · MADISON, ALABAMA

AGENDA

Three hours. Finish the box, build the power, ready to smoke-test.

0:00	Welcome & Safety Brief	15 min
0:15	Where the Board Stands	15 min
0:30	Finish the Control-Box Wiring	60 min
1:30	Break	10 min
1:40	ROV Power Wire + Tether Connection	45 min
2:25	Fuse Selection from Measured Full-Load Amps	15 min
2:40	Pre-Power-Up Meter-Check (Three Reads × Every Pad)	20 min
3:00	Wrap-Up & Homework	0 min

We do NOT power the board up today. Smoke test waits for 1.7's pre-pool review.

TODAY'S ROLES

Four jobs. Four students. Different every week.

Every meeting we assign these four — by season's end, every student has done every job.

NOTETAKER / SCRIBE

TODAY: _____

Captures every decision, action item, and blocker. Files notes with the CFO for the engineering notebook.

SAFETY BRIEFER

TODAY: _____

Delivers the standing four-question safety brief at the start of the meeting.

PHOTOGRAPHER

TODAY: _____

Captures photos and short videos throughout — feeds the engineering notebook and the marketing display.

TIMEKEEPER

TODAY: _____

Holds the meeting to the agenda's clock. Calls the time at each block change.

SECTION 01

Welcome & Safety Brief

0:00 - 0:15 · 15 MIN

Same ritual every meeting. We get sharper at it; the kids start running it.



Every meeting opens with a safety brief.

Why we do this

Real engineering and dive teams brief safety before every job — not because they expect to fail, but because the discipline of asking the same four questions every time is what makes a team safe in the long run. We start the habit now, when the stakes are small, so it's automatic when they aren't.

- 1 **What are we doing today?**
- 2 **What could hurt someone?**
- 3 **How do we prevent it?**
- 4 **Who do we tell if something happens?**

TODAY'S BRIEF · MEETING 6

THE WORK

Finish soldering the TriggerFish control box. Build the ROV power wire from battery to board. Terminate the tether at the control box.

THE HAZARDS

Same as 1.5 plus stripped-wire ends (sharp) and 12V DC at the power-wire end once the battery is connected. Battery is OFF during build.

PREVENTION

Battery stays disconnected until pre-power-up review. No exceptions. Same iron, glasses, fume, hand-wash rules as 1.5.

IF SOMETHING HAPPENS

Stop. Disconnect the battery first (if connected). Tell the advisor. Burns get cool water for 15 minutes; cuts get the first-aid kit.

LOOKING AHEAD

1.7 is the capsule build and first pool day. Pool prerequisites — lifeguard, swim checks, photo releases — get confirmed this week, not at the pool deck.

SECTION 02

Where the Board Stands

0:15 - 0:30 · 15 MIN

Inventory what 1.5 closed. Name the open joints. Read the next block from the Guide.



Where 1.5 left us — and what's left to do.

Coming out of 1.5 the board carries the fuse holder, the two 1000 μ F / 16V caps, the 3 A practice fuse, and the next block of components per the SeaMATE Guide. Every completed joint should be meter-checked from last week's discipline pass.

Tonight we close the remaining components on the control box itself, build the ROV power wire from battery to board, terminate the tether at the control box, and re-fuse the circuit to the measured full-load amps from the watt meter (the 3 A practice fuse comes out).

We do NOT power the board up tonight. The smoke test waits for 1.7's pre-pool review. The reason is simple: every wiring fault we catch dry is one we don't catch wet.

SECTION 03

Finish the Control-Box Wiring

0:30 - 1:30 · 60 MIN

Sixty minutes hands on. Same build loop as 1.5: read, stage, solder, inspect, meter.



Read the slide. Stage. Solder. Inspect. Meter. Next.

Same loop as 1.5. We don't get ahead of the Guide.

1

Read the slide as a team

One reader, others listen. No iron picked up until the team agrees on what the slide is asking for.

2

Stage the components

Lay them out next to the board, oriented the way they'll go in. Polarity-sensitive parts marked clearly with a piece of tape or a label.

3

Solder one at a time

Whoever has the iron owns that joint. Others coach, photograph, or cue up the next component.

4

Inspect and meter-check

Shiny, smooth, concave. Continuity to the next pad in the circuit; no shorts to ground or neighboring pads.

5

Move to the next slide

Only when the current step passes inspection AND meter-check. The discipline is what gets us through inspection in April.

BREAK

Ten minutes.

We resume in ten — and, like always, we start on time.

SECTION 04

Power Wire + Tether

1:40 - 2:25 · 45 MIN

From battery to board. From board to tether. Strain relief at every termination.



Battery to board. Three things matter.

GAUGE

Heavy enough to carry the measured full-load amps without voltage drop. The team's spec for the TriggerFish: 14 AWG silicone-jacketed, two conductors red/black, ring terminals at the battery end.

POLARITY

Red to positive, black to negative — at BOTH ends. Marked clearly. Reversed polarity at the board feeds the wrong rail and ends the night. Two people verify before commit.

STRAIN RELIEF

Heatshrink + cable tie at the board end. The wire must not be the load-bearing member if someone trips over it. Tape and zip ties as primary strain relief fail safety inspection in April.

Pre-terminated. Strain relief at both ends.

CONTROL BOX END

Tether conductors terminate at the control box per the SeaMATE Guide. Each conductor is meter-checked end-to-end before the heatshrink goes on the splice.

ROV END

The other end gets terminated to the thrusters and cameras in 1.8. Tonight we land the control-box end and verify continuity through every conductor.

SECTION 05

The Fuse Lives by Measurement

2:25 - 2:40 · 15 MIN

*Watt meter reads the full-load amps. The fuse sizes one step above.
That number lives in the engineering notebook.*



Measure. Round up to the next blade size. Document.

MATE safety inspection asks for the FLA math. The team that has it cited from a measurement passes clean.

1

Measure full-load amps with the watt meter

Battery connected ONLY to the watt meter (not the board). All thrusters and electronics at simulated full-load if possible; if not, the highest measured draw from 1.5 / 1.6 stages plus a safety margin.

2

Round up to the next standard blade-fuse size

If measured FLA is 4.2 A, fuse at 5 A. If 7.1 A, fuse at 10 A. Standard blade sizes — 3 / 5 / 7.5 / 10 / 15 / 20 A. The 3 A practice fuse comes out tonight.

3

Document in the engineering notebook

Measured FLA. Fuse size selected. Citation: ELEC-NRD-003 (the MATE rule that requires the fuse to be sized from measured draw, not guessed).

4

Install and verify the new fuse seats cleanly

New fuse in the holder. Multimeter continuity through the holder. Photograph the holder + the engineering-notebook entry side by side.

SECTION 06

Pre-Power-Up Meter-Check

2:40 - 3:00 • 20 MIN

Every pad. Three reads. Sign the board off before we leave.



If we miss a short tonight, we find it the loud way next week.

- 1 Continuity from every pad to its intended next pad**
Walk the schematic with the multimeter. Every connection on the slide gets a beep at the bench. Silence means re-route or re-flow.
- 2 No shorts to ground or to neighboring pads**
Probe every pad against ground and against immediate neighbors. Beep here means a solder bridge — fix it tonight.
- 3 Polarity confirmed at the power-wire termination**
Red to positive rail. Black to negative rail. Eyes on the board, eyes on the Guide, eyes on the wire. Three references; one orientation.
- 4 Sign the board off in the engineering notebook**
Whoever did the final meter-check signs the page. Date, time, results. That signature is the answer when a judge asks 'who verified this?' in April.

SECTION 07

Wrap-Up & Homework

3:00 - 3:00 • 0 MIN

Capture what we decided. Close the loop. Hand off the next move.



WRAP-UP & HOMEWORK

Before we meet again.

EVERY STUDENT

- 1 Pool prerequisites for 1.7 — confirm with your parents that your photo-release form is signed and your swim-comfort form is on file. We do not get in the water without both.
- 2 Re-read the SeaMATE Guide sections on capsule construction. The kit doesn't come with capsules — we build them in 1.7.
- 3 Bring closed-toe shoes AND a towel + change of clothes next week. The capsule drop is on a pool deck.

THE PARALLEL TRACK · LEADS & ADVISORS

Advisors: confirm lifeguard coverage, host pool rules in writing, photo releases returned for every kid expected in frame. These are the gating items for 1.7 — without any one of them, the pool day slips a week.

NEXT MEETING · WEEK 7

Capsule Build & First Splash

Build the watertight capsules — PVC, end caps, O-rings, marine epoxy. Pool deck. Every kid in the water. Capsule drop #1: assemble dry, drop wet, retrieve, check the towel inside, log the result.

ACTION ITEMS

What we decided. Who owns it. When it's due.

The Notetaker fills these in during the meeting and reads them back before we leave.

ITEM	OWNER	DUE

Tools accounted for. Shop reset.

We do this together, every meeting. No exceptions — the Tool Steward leads.

- ☐ Soldering irons unplugged, cool, and put away.
- ☐ Control box secured in its tray; power wire and tether coiled neatly beside it.
- ☐ Battery STILL DISCONNECTED — confirmed by two people before leaving the room.
- ☐ Multimeter and watt meter back in their cases.
- ☐ Workshop swept; benches wiped; sharps disposed.
- ☐ Today's photos uploaded to the team folder.
- ☐ Notetaker's meeting notes filed with the CFO for the engineering notebook.

THE BOARD IS DONE. THE BOX IS READY.

Next week, the capsules — and the pool.

Smoke test happens during 1.7's pre-pool review. If the meter-check holds, we power up and we drop the first capsule the same day.